AMAL JYOTHI
COLLEGE OF ENGINEERING
AUTONOMOUS
KANJIRAPPALLY

ASSIGNMENT-1

23INMCA304 – Fundamentals of Data Science

TITLE: Framing Real-World Problems Using Big Data Concepts

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Return and refund pattern analytics

1. Domain and Organizational Context

Domain: Retail (Grocery and General Merchandise Supermarket Chain)

Organizational Context: Reliance Smart is a major supermarket chain under Reliance Retail. It sells daily groceries (fruits, vegetables, dairy, packaged foods), household items, apparel, and small electronics. Many stores are in Kerala (Thiruvananthapuram, Kochi, etc.) and across India. It combines physical stores with online shopping through JioMart for quick delivery.

Scale of Operations: National – thousands of Reliance Smart, Smart Point, and Smart Bazaar stores serving crores of customers. High sales during festivals like Onam, Diwali, and year-end. Handles lakhs of transactions daily, with growing online orders via JioMart (recently reaching 1.6 million daily orders in quick commerce).

Key Stakeholders Involved:

- Customers (in-store and online shoppers for daily needs)
- Store and warehouse staff
- Delivery partners for quick commerce
- Suppliers and farmers (for fresh produce)
- Company management (for planning and profits)
- Government regulators (food safety, consumer complaints)

2. Problem Situation

The main problem is high volume of product returns, exchanges, and quality complaints, causing big financial losses, food waste, and unhappy customers.

In grocery retail like Reliance Smart, people return spoiled vegetables/fruits, wrong-packed items, or defective goods (especially from JioMart online). Return rates are high for perishables during rainy seasons or festivals. This leads to:

- Refund payouts and reverse delivery costs
- Wasted stock (thrown away spoiled items)
- Extra work for staff

The problem grows bigger with JioMart's fast growth (quick 30-60 minute deliveries) and more online impulse buys. Current basic POS systems and manual checks are slow and cannot handle thousands of daily claims. This causes delays in refunds, more complaints, and customers switching to competitors like DMart or local shops.

3. Data Generation Environment

Multiple Data Sources Involved:

- In-store: Barcode scans, loyalty card swipes, billing data
- Online/JioMart: App orders, delivery GPS tracking, feedback
- Returns: Customer reasons, photos of returned items
- Inventory: Warehouse stock updates, supplier records
- External: Weather data (affects produce freshness), festival trends

Types of Data:

- Structured: Product codes, prices, quantities, timestamps, customer phone numbers
- Semi-structured: App logs and delivery status (JSON format)
- Unstructured: Complaint text ("tomatoes spoiled"), photos/videos of items, support chats

How Data is Generated:

- Real-time/event-driven: Every scan, order placement, or return request happens instantly
- Periodic: Daily stock reports from stores/warehouses
- Continuous: Temperature sensors in cold storage, live app sessions

4. Decision-Making Challenges

Huge data volume, speed, and variety make quick and correct decisions hard. Examples:

- Difficult to spot patterns like high returns on certain veggies after rain
- Manual checks for fake returns (swapped items) take too long
- Cannot predict stock needs fast, leading to over-buying (waste) or shortages

Who is Affected:

- Customers: Slow refunds, poor quality → lose trust
- Store staff: Extra work on returns instead of helping others
- Company: Money loss on waste/refunds, lower profits
- Suppliers: More complaints, fewer repeat orders

5. Justification for Big Data Orientation

Traditional databases and simple retail software are not enough for Reliance Smart's scale. They handle small data but slow down with crores of transactions, cannot mix numbers with photos/text, and take hours to find return patterns.

Big Data technologies are needed because:

- Handle extreme volume from daily in-store + online orders across thousands of stores
- Process high velocity for real-time checks (flag bad batches or fraud during return)
- Manage variety – combine sales data, images, and complaints for full picture
- Check veracity (clean noisy/fake data like wrong reasons)
- Deliver value – predict returns early, improve supplier quality, personalize offers to reduce complaints, cut waste, and increase profits

This shift helps Reliance Smart give better service, save costs, keep customers loyal, and stay strong against quick commerce rivals.